



COURSE SLIDES

Agentic AI for Finance

Course Code: C057

Conducted by Tertiary Infotech Academy Pte Ltd · UEN 201200696W

Version v1.0 · 29 July 2026

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COURSE ADMINISTRATION

Welcome & Housekeeping

Course Trainer Profiles

1

Assigned Tertiary Infotech Academy Trainer

Finance-automation and agentic-AI facilitator who guides control-first design, deterministic verification, evidence-backed analysis and hands-on application to the synthetic Northstar Components scenario.

Let's Know Each Other

1 Your name and organisation / role.

2 Your experience with this subject (if any).

3 What you want to be able to do after this course.

Ground Rules

1

Set your mobile phone to silent mode.

2

Participate actively — no question is too small.

3

Mutual respect: agree to disagree.

4

One conversation at a time.

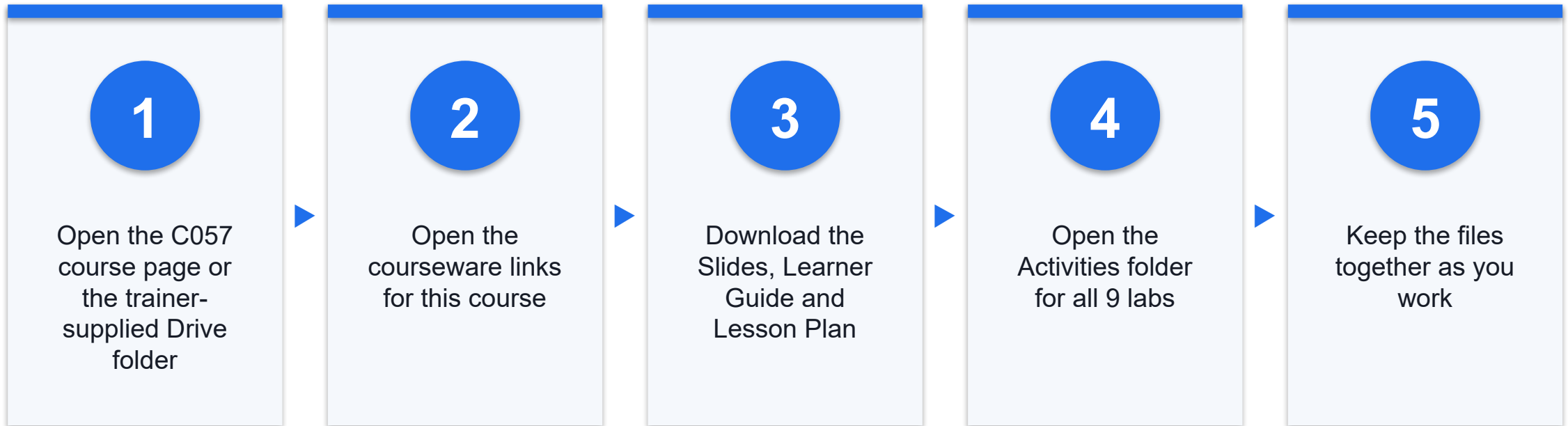
5

Be punctual; return from breaks on time.

6

Learn by doing — try every hands-on lab.

Access Course Materials



Course page: <https://www.tertiarycourses.com.sg/agent-ai-for-finance.html>

Lesson Plan — 2 Days, 15 instructional hours

Course sequence

- Day 1 — Design and Automate — build controlled finance-agent foundations and workflows
 - Topic 1: Getting Started with Agentic AI for Finance (Labs 1–2)
 - Topic 2: Automating Financial Workflows with AI Agents (Labs 3–5)

Day 2

- Day 2 — Analyse and Govern — produce evidence-backed decisions and a deployable operating model
 - Topic 3: Analysis and Insights with AI Agents (Labs 6–7)
 - Topic 4: Deploying and Governing Financial AI Agents (Labs 8–9)
- Daily timing
 - 9:30 am–6:30 pm (1-hour lunch; two 15-minute tea breaks; 7.5 instructional hours)

Learning Outcomes

1

Design the Agent

Design a bounded finance agent with clear instructions, approved data connections, tool permissions and human review points.

2

Automate Workflows

Automate reporting, reconciliation, forecasting and invoice workflows while preserving deterministic checks and human approval.

3

Analyse & Explain

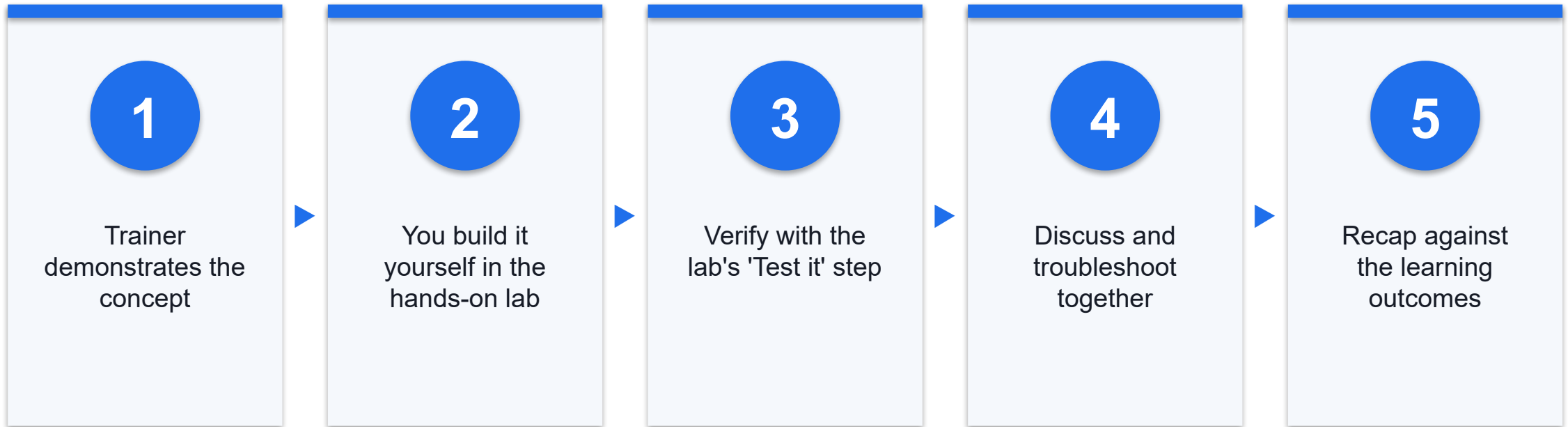
Produce traceable financial analysis, scenarios, recommendations and visualisations from verified data.

4

Deploy & Govern

Deploy and govern finance agents with security, auditability, evaluation, monitoring and scalable operating controls.

How You'll Learn





CORE CONCEPTS

Course Fundamentals

The Controlled Finance-Agent System

1

Ground

Use approved, versioned finance evidence with explicit grain, period, units and ownership.

2

Calculate

Keep arithmetic, matching, thresholds and write rules deterministic and independently testable.

3

Reason

Use the model to plan, classify exceptions, challenge assumptions and draft explanations.

4

Approve

Route material decisions and irreversible actions to an authorised human with visible evidence.

One Agent Run

1

Receive

Validate goal, identity, inputs and permission.

2

Plan

Select the next bounded step and approved tool.

3

Act

Read, calculate, classify or draft within limits.

4

Observe

Check tool result, totals, exceptions and stop conditions.

5

Review

Approve, correct, escalate or close with a complete run record.

COURSE OPERATING PRINCIPLE

Let the model manage ambiguity; let controls own financial truth.

Every material number is sourced or calculated, every exception is owned and every consequential action is approved.

What You Will Build

1

Agent Foundation

use-case and tool-risk register · finance prompt and data contracts

2

Workflow Portfolio

reconciliation and close brief · forecast and invoice exception routes

3

Decision Pack

verified financial analysis · scenario insight and chart specification

4

Operating Model

security and evidence controls · evaluation, monitoring and deployment plan

How Every Lab Progresses

1

Open the approved synthetic Northstar Finance checkpoint.

2

Apply a bounded prompt or deterministic finance rule.

3

Separate source facts, calculations, model judgement and human decisions.

4

Test the output against exact control totals or observable routing rules.

5

Save the run evidence and the next rejoin checkpoint.

Agent, Assistant or Automation?

1

Assistant

A person chooses each request and decides what to do with the answer.

2

Automation

Fixed rules execute a known sequence with predictable branches.

3

Agent

A model chooses among approved next actions inside a bounded workflow.

4

Control rule

Use the least autonomous pattern that meets the business need.

The Finance Truth Stack

1

System of record

Authoritative ledger, document or approved business source.

2

Deterministic layer

Queries, formulas, matching rules, thresholds and validations.

3

Model layer

Planning, classification, drafting, critique and exception explanation.

4

Decision layer

Named human authority, rationale and controlled record update.

The Four Permission Questions

1

Read what?

Specific data classes, rows, fields, periods and documents.

2

Act how?

Allowed tools, parameters, destinations and financial limits.

3

Stop when?

Completion, uncertainty, failure, budget and risk thresholds.

4

Who approves?

Role, evidence, segregation, response time and fallback.

TOPIC 01

Getting Started with Agentic AI for Finance

agent foundations · tools and platforms · finance prompting · safe data connections

Key Concepts — Getting Started with Agentic AI for Finance

1

Agentic workflow

A model uses instructions and tools in a bounded loop to pursue a goal, observe results and decide the next step.

2

Finance control boundary

Define permitted data, actions, limits, owners, stop conditions and evidence before the first run.

3

Tool architecture

Separate read tools, calculation tools and action tools; grant only the access required for the task.

4

Finance prompt contract

State the source boundary, method, output schema, acceptance checks, uncertainty rules and reviewer.

5

Data contract

Document grain, fields, units, period, ownership, quality checks, lineage and refresh time.

6

Deterministic truth

Use formulas and source-system rules for arithmetic, matching and posting; use the model for language and exceptions.

Introduction to Agentic AI in Finance

WHAT

- An AI agent is a system in which a model controls part of a multi-step workflow, selects approved tools, observes results and continues until it reaches a defined completion or stop condition. A chatbot answers a turn; a deterministic automation follows fixed rules; an agent chooses among bounded next actions.

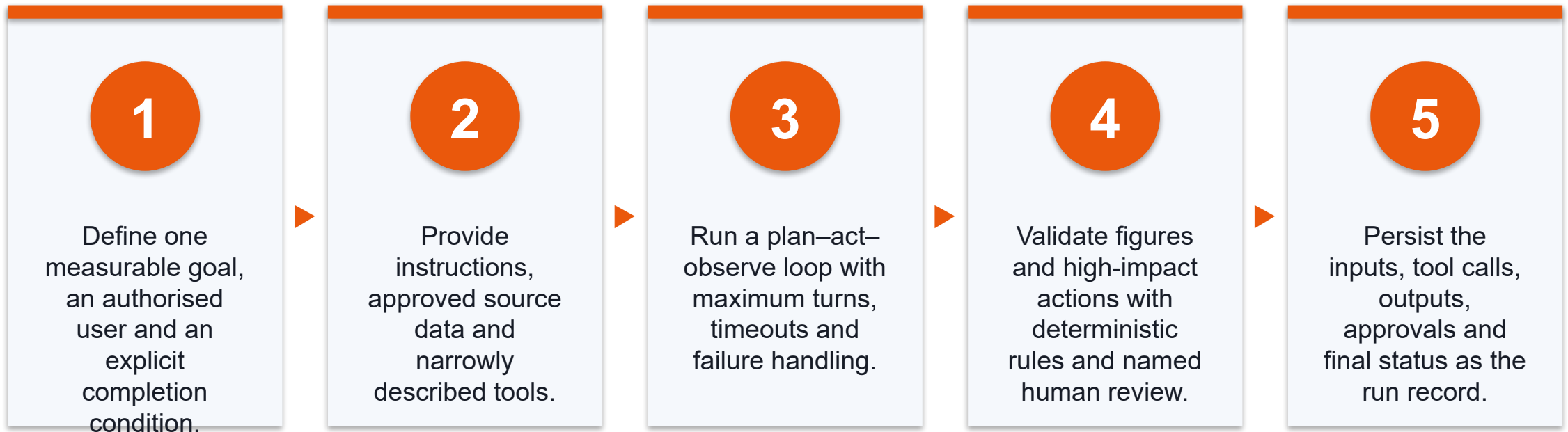
WHY IT MATTERS

- Finance work mixes repeatable calculations with ambiguous exceptions and narrative judgement. Agents can coordinate those steps, but fluent output must never replace ledgers, policies, formulas or accountable decisions. The useful design question is not how autonomous the system can be, but which decisions it may make safely.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Introduction to Agentic AI in Finance



In Practice — Introduction to Agentic AI in Finance

WORKED EXAMPLE

- Goal: prepare a draft month-end variance brief from the approved June extract.
- The agent reads the extract, invokes a calculation step, requests clarification for an unmapped account and drafts commentary.
- The finance manager checks the figures and approves the brief; the agent has no permission to post a journal or send externally.

DECISION GUIDE

- **Use when**
 - The workflow has multiple steps, exceptions or unstructured inputs that fixed rules alone handle poorly.
 - A reliable source boundary, measurable checks and a human owner can be defined.
- **Avoid when**
 - A spreadsheet formula, database query or fixed rule completes the task more simply and predictably.
 - The agent would make an irreversible financial commitment without an authorised approval gate.

Quality Lens — Introduction to Agentic AI in Finance

Bounded

- Purpose, inputs, tools, limits, exit conditions and fallback are explicit.

Grounded

- Every material figure traces to an approved source or deterministic calculation.

Owned

- A named role reviews exceptions and remains accountable for the final decision.

Popular AI Agent Tools and Platforms

WHAT

- Agent platforms range from browser assistants and configurable workspace agents to low-code workflow builders and software development kits. All combine a model, instructions and tools; they differ in integration depth, observability, deployment control, cost and the skill needed to operate them.

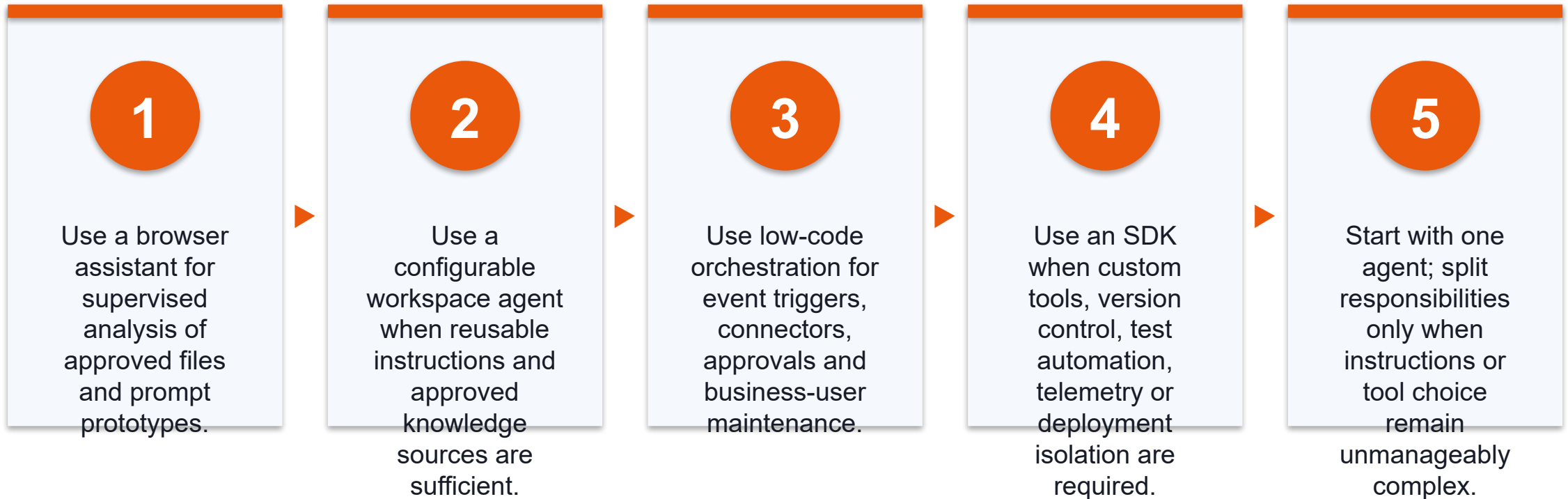
WHY IT MATTERS

- Choosing a platform before defining the workflow often produces an expensive demonstration with weak controls. Finance teams should first classify the data, actions and evidence required, then select the least complex platform that can satisfy security, integration, evaluation and operating needs.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Popular AI Agent Tools and Platforms



In Practice — Popular AI Agent Tools and Platforms

WORKED EXAMPLE

- A finance analyst prototypes variance commentary in an approved browser assistant with a synthetic CSV.
- A process owner then maps the same prompt contract into a low-code flow with a read-only data connector and manager approval.
- The engineering team uses an SDK only when the workflow needs custom identity, detailed tracing and controlled release gates.

DECISION GUIDE

- **Use when**
 - Comparing implementation options against an already defined finance workflow and control matrix.
 - A proof of value needs a clear path from supervised prototype to monitored operation.
- **Avoid when**
 - Selecting a platform because it has the most connectors or appears most autonomous.
 - Allowing a connector to inherit broad user permissions without a tool-by-tool risk review.

Quality Lens — Popular AI Agent Tools and Platforms

Fit

- Workflow complexity and control requirements drive the platform choice.

Visibility

- Runs, tool calls, versions, failures and approvals can be inspected.

Portability

- Instructions, schemas, tests and data contracts are not trapped in one interface.

Writing Effective Prompts for Finance

WHAT

- A finance prompt is an operating contract for a model. It identifies the role and objective, delimits authorised evidence, specifies the method and tools, defines a machine-checkable output, states acceptance criteria and tells the model when to ask, stop or escalate.

WHY IT MATTERS

- A vague request such as 'analyse the numbers' invites inconsistent calculations, unsupported causes and hidden assumptions. A structured prompt makes the evidence boundary and review process observable, so a useful draft can be reproduced and tested.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Writing Effective Prompts for Finance



In Practice — Writing Effective Prompts for Finance

WORKED EXAMPLE

- The prompt names the June close, approved files and SGD units, then supplies the exact variance formula.
- The output schema requires Account, Actual, Budget, Variance, Direction, Evidence and Reviewer_note.
- An account without a budget mapping is returned as UNKNOWN and routed to the controller instead of being invented.

DECISION GUIDE

- **Use when**
 - A model must transform, explain, classify or critique financial information in a repeatable format.
 - The output will be checked against source evidence and deterministic acceptance rules.
- **Avoid when**
 - The prompt contains secrets, personal data or restricted information not approved for the service.
 - The model is asked to calculate totals without a deterministic reconciliation step.

Quality Lens — Writing Effective Prompts for Finance

Specific

- Task, scope, units, period, schema and thresholds are explicit.

Testable

- Acceptance checks can be evaluated without interpreting the prose.

Fail-safe

- Missing evidence triggers UNKNOWN, clarification or escalation.

Connecting Agents to Financial Data

WHAT

- A financial data connection is a governed interface to a defined dataset, not unrestricted access to a drive or ledger. Its contract records source owner, grain, schema, units, accounting period, refresh time, permitted use, quality rules, lineage and access mode.

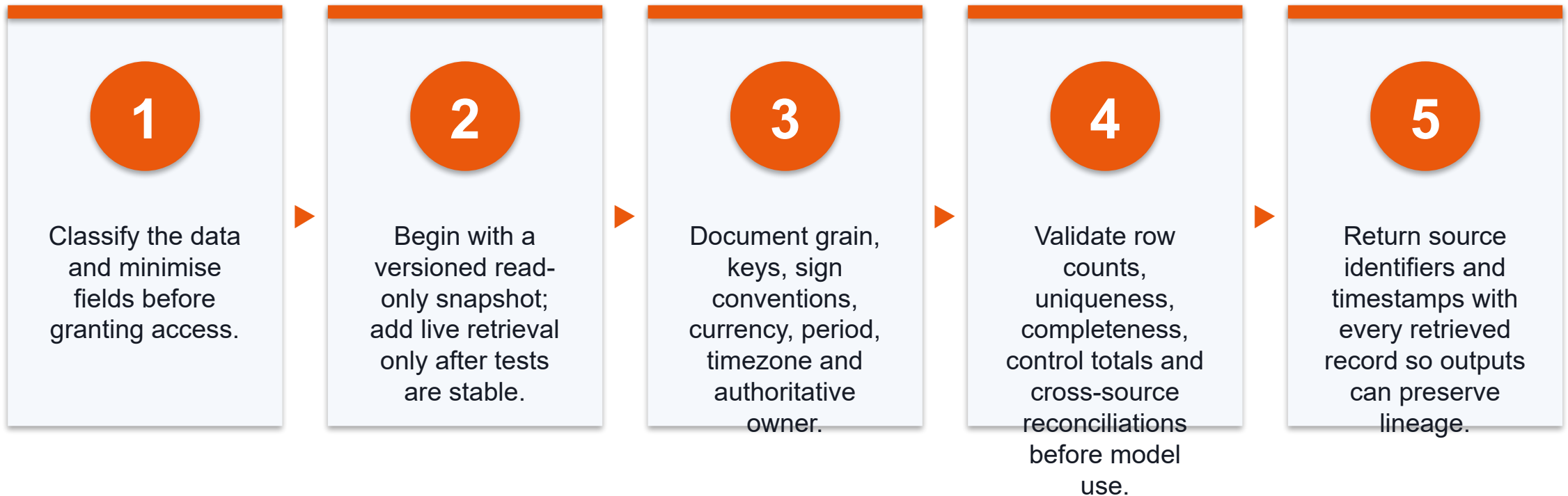
WHY IT MATTERS

- Many apparent model failures are data failures: duplicate rows, mixed periods, stale extracts, mismatched currencies or unclear sign conventions. A read-only snapshot and manifest make the analysis reproducible and reduce the blast radius of prompt injection, accidental writes and over-broad access.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Connecting Agents to Financial Data



In Practice — Connecting Agents to Financial Data

WORKED EXAMPLE

- The agent receives a June general-ledger extract and a budget table through separate read-only tools.
- A manifest records 30 June cut-off, SGD units, account grain and control totals; duplicate Journal_ID values fail ingestion.
- The prompt references source labels rather than copying an entire finance drive into context.

DECISION GUIDE

- **Use when**
 - The agent needs repeatable access to structured finance data or approved documents.
 - Data owners can define quality checks, permissions, retention and lineage.
- **Avoid when**
 - The connector exposes unrelated folders, credentials or write permissions.
 - The source has no stable key, period, unit or owner and cannot be reconciled.

Quality Lens — Connecting Agents to Financial Data

Minimal

- Only necessary rows, fields and periods are exposed.

Reconciled

- Control totals and key constraints pass before analysis.

Traceable

- Every output can identify its source, version and retrieval time.

Hands-On Labs — Getting Started with Agentic AI for Finance

Lab 1

- Design the Finance Agent Charter and Prompt Contract

Lab 2

- Connect and Profile Approved Financial Data

Design the Finance Agent Charter and Prompt Contract

LAB 1

You will select the monthly-close reporting use case from the supplied register and define what the agent may read, calculate and draft. You will separate deterministic controls from model judgement, specify stop conditions and create a reusable prompt contract that preserves finance ownership.

You'll build

01-foundation/finance-agent-charter.md, tool-risk-register.csv, prompt-contract.md and run-evidence/L01-tabletop-run.md for a read-only June close-report agent.

1 Select the use case

2 Bound the agent

3 Rate the tools

4 Write the prompt

Tools: Spreadsheet · text editor · approved AI assistant · agent-use-case-register.csv · data-dictionary.csv

Design the Finance Agent Charter and Prompt Contract

Test it

The charter must contain all 13 headings and one measurable completion condition. The tool register must contain exactly five Tool_ID rows, with PUBLISH_BRIEF prohibited and every non-read action assigned a human gate. The prompt must contain all five C-L-E-A-R sections and require Source_ID, FACT/HYPOTHESIS/UNKNOWN separation and a reviewer checklist. Both tabletop scenarios must stop with no invented data and no write action.

Connect and Profile Approved Financial Data

LAB 2

You will connect the foundation to three synthetic snapshots: cash ledger, bank statement and budget-versus-actual. Before any model sees the data, you will document grain and sign conventions, verify keys and control totals, remove unneeded fields and test whether the assistant cites stable Source_ID values.

You'll build

01-foundation/source-manifest.csv, data-contracts.md, data-quality-report.md, approved-source-package.md and run-evidence/L02-grounding-test.md.

1 Copy snapshots

2 Record lineage

3 Run quality checks

4 Minimise access

Tools: Spreadsheet · text editor · approved AI assistant · cash-ledger.csv · bank-statement.csv · budget-actual.csv

Connect and Profile Approved Financial Data

Test it

The manifest must contain exactly three Source_ID rows and the expected counts and totals. Each data contract must contain all 13 required fields. The quality report must show a result for every stated check. The grounding test must return seven correct values with seven valid Source_ID citations and zero unsupported fields. The tool register must restrict BANK_JUN to reconciliation.

Recap — Getting Started with Agentic AI for Finance

1

You can now: LO1: design a bounded finance agent with instructions, tools, limits and human review

2

You can now: LO1: connect an agent to approved financial data with quality, lineage and least-privilege controls

TOPIC 02

Automating Financial Workflows with AI Agents

reporting and reconciliation · forecasting and planning · invoice processing · human-in-the-loop controls

Key Concepts — Automating Financial Workflows with AI Agents

1

Control-first workflow

Map source, calculation, exception, approval, record update and evidence before adding a model.

2

Reconciliation

Match records by approved keys and tolerances, explain only the remaining exception set and preserve adjusted balances.

3

Driver-based forecast

Translate explicit volume, price, rate and cost assumptions into formulas and comparable scenarios.

4

Document pipeline

Capture, extract, validate, match, route, review and record without letting extraction confidence authorise payment.

5

Human gate

Route material, unusual, low-confidence or irreversible actions to an authorised role.

6

Idempotent action

A repeated run must not create duplicate records, payments or notifications.

Automating Reporting and Reconciliation

WHAT

- A reporting agent assembles verified calculations and narrative from approved sources. A reconciliation compares two records of the same economic activity, applies exact or authorised tolerant matches, isolates exceptions and proves that adjusted balances agree.

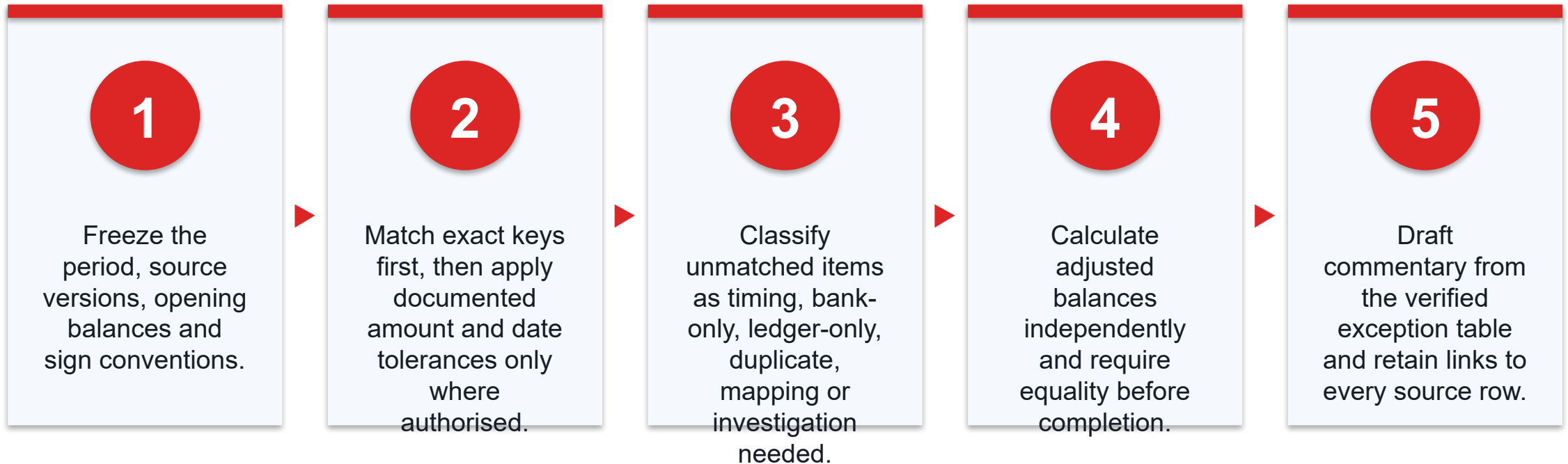
WHY IT MATTERS

- Language models are useful for classifying exception descriptions and drafting commentary, but matching logic and control totals should remain deterministic. Combining both prevents a persuasive narrative from hiding an unreconciled difference.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Automating Reporting and Reconciliation



In Practice — Automating Reporting and Reconciliation

WORKED EXAMPLE

- Seven bank and ledger transactions match by reference and amount.
- One outstanding payment reduces the adjusted bank balance; a bank fee and interest item adjust the ledger.
- Both adjusted balances equal SGD 64,680, so the agent drafts a close note and routes the bank-only items for posting review.

DECISION GUIDE

- **Use when**
 - The workflow has stable matching keys, tolerances and a defined exception owner.
 - Narrative is generated only after control totals and adjusted balances pass.
- **Avoid when**
 - The model is allowed to 'make the balances agree' by deleting or inventing transactions.
 - Matching rules, cut-off or sign conventions are undocumented.

Quality Lens — Automating Reporting and Reconciliation

Complete

- Every source row is matched once or appears in an owned exception queue.

Balanced

- Independent adjusted balances agree exactly or within an approved tolerance.

Explainable

- Each match rule and exception classification is visible.

Building Forecasting and Planning Agents

WHAT

- A forecasting agent converts explicit business drivers and assumptions into a time-phased projection, compares scenarios, explains sensitivities and records who approved each assumption. The numerical engine is deterministic; the model supports assumption discovery, challenge and communication.

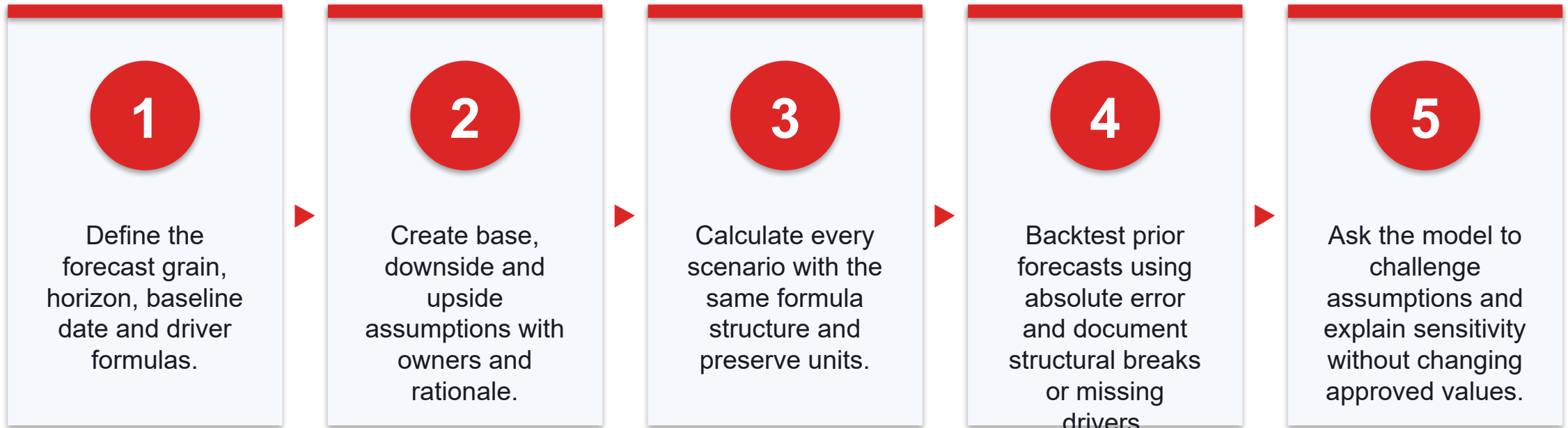
WHY IT MATTERS

- A single forecast can create false precision. Driver-based scenarios expose how revenue growth, cost rates and operating expenses affect outcomes. Backtesting and assumption ownership make forecast error a learning signal rather than a reason to rewrite history.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Building Forecasting and Planning Agents



In Practice — Building Forecasting and Planning Agents

WORKED EXAMPLE

- June revenue of SGD 125,000 is the base; July base growth is 3%, cost of goods is 56% of revenue and operating expense is SGD 40,000.
- The deterministic July operating-profit calculation is SGD 16,650.
- The agent compares downside and upside cases, names the largest driver and presents the assumption owner and review date.

DECISION GUIDE

- **Use when**
 - The organisation can identify controllable drivers and maintain scenario assumptions.
 - Decision-makers need ranges, sensitivities and triggers rather than one unsupported point estimate.
- **Avoid when**
 - The agent invents future events, probabilities or market data that were not supplied.
 - Forecast outputs are written back as an approved plan without owner review.

Quality Lens — Building Forecasting and Planning Agents

Formula-led

- Scenarios share transparent formulas and units.

Assumption-owned

- Every material driver has a source, owner and review trigger.

Backtested

- Error is measured on prior periods and feeds the next revision.

Document and Invoice Processing

WHAT

- An invoice agent captures a document, extracts fields, validates the supplier and invoice identity, checks purchase and receipt evidence, applies tolerances, routes exceptions and creates a draft record for approval. Extraction is evidence capture, not authority to post or pay.

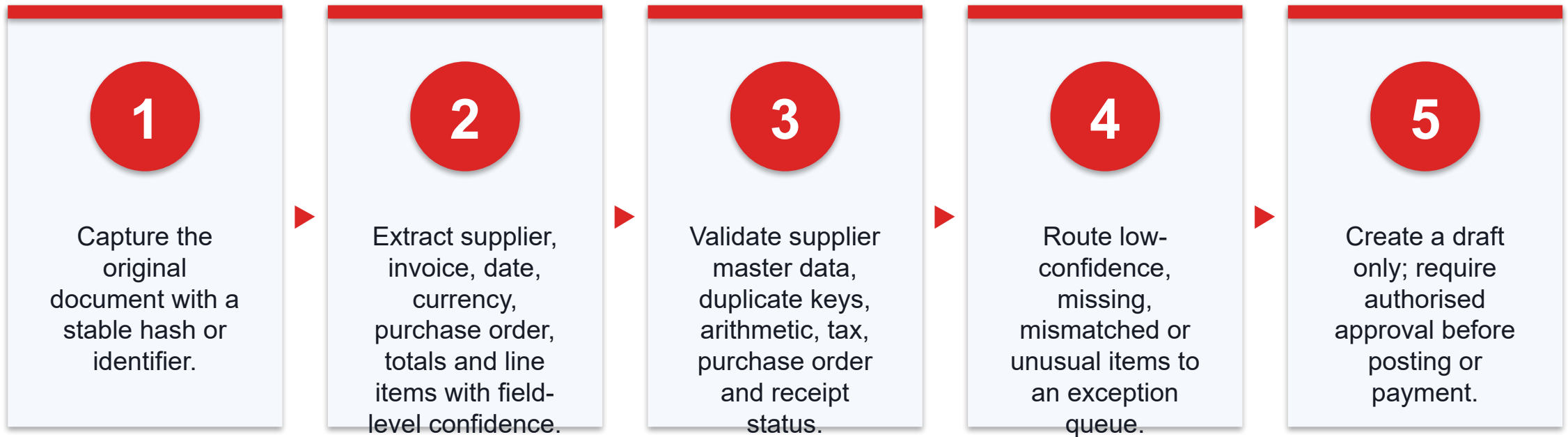
WHY IT MATTERS

- Invoices are semi-structured and arrive in many formats, making extraction a good AI use case. Financial commitment, duplicate prevention, tax treatment and payment remain high-impact controls that require deterministic validation and explicit human authority.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Document and Invoice Processing



In Practice — Document and Invoice Processing

WORKED EXAMPLE

- An invoice total matches its purchase order and receipt, but another record exceeds the amount tolerance by SGD 400.
- A missing purchase order, a duplicate flag and a low-confidence total each take different exception routes.
- The workflow preserves the source document and reviewer correction so extraction performance can be improved.

DECISION GUIDE

- **Use when**
 - Documents follow known business processes and reviewers can correct exceptions efficiently.
 - The system can preserve originals, extracted fields, confidence, validation results and approvals.
- **Avoid when**
 - The agent may create a supplier, change bank details, post or pay without independent verification.
 - Confidence score alone is treated as proof that the transaction is valid.

Quality Lens — Document and Invoice Processing

Source-preserved

- Original document and extracted fields remain linked.

Three-way checked

- Invoice, purchase order and receipt are compared where applicable.

Draft-only

- Irreversible actions remain behind authorised approval.

Human-in-the-Loop Controls

WHAT

- Human-in-the-loop control inserts a named person at a defined decision point to review evidence, correct the record, approve an action or take over a failed run. It is a designed operating role with thresholds and service levels, not a generic instruction to 'check the output'.

WHY IT MATTERS

- Finance actions vary sharply in impact and reversibility. A read-only draft can tolerate more automation than a journal, supplier-master change or payment. Risk-tiered gates protect segregation of duties while still allowing low-risk work to move quickly.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Human-in-the-Loop Controls



In Practice — Human-in-the-Loop Controls

WORKED EXAMPLE

- Read-only variance drafting proceeds automatically after reconciled totals pass.
- An invoice within tolerance creates a draft, while a missing purchase order or bank-detail change is held for independent review.
- A retry uses the same idempotency key, so the workflow cannot create a second draft invoice.

DECISION GUIDE

- **Use when**
 - A task is valuable to automate but exceptions or actions can create material impact.
 - Authorised reviewers, thresholds, evidence and response times can be assigned.
- **Avoid when**
 - The reviewer sees only a recommendation and cannot inspect evidence or change the outcome.
 - Approval is performed by the same identity that initiated a restricted action without segregation.

Quality Lens — Human-in-the-Loop Controls

Risk-tiered

- Review strength rises with sensitivity, impact and irreversibility.

Actionable

- The reviewer has evidence, choices, reason codes and a deadline.

Independent

- Segregation of duties is maintained for restricted actions.

Hands-On Labs — Automating Financial Workflows with AI Agents

Lab 3

- Build the Reconciliation and Close-Reporting Agent

Lab 4

- Build the Driver-Based Forecasting Agent

Lab 5

- Design the Invoice Exception and Human-Review Agent

Build the Reconciliation and Close-Reporting Agent

LAB 3

You will use deterministic matching before asking the model to classify or explain anything. The completed reconciliation must account for every source row, prove equal adjusted balances and feed only verified exceptions and budget variances into the close-report prompt.

You'll build

02-automation/reconciliation-june.csv, exception-queue.csv, adjusted-balance-proof.md, june-close-brief.md and run-evidence/L03-reconciliation-run.md.

1 Match exact records

2 Classify exceptions

3 Prove adjusted balances

4 Calculate variances

Tools: Spreadsheet · text editor · approved AI assistant · CASH_LEDGER_JUN · BANK_JUN · BUDGET_ACTUAL_JUN

Build the Reconciliation and Close-Reporting Agent

Test it

The reconciliation must contain seven one-to-one matched pairs and the exception queue exactly three source rows. Ledger ending balance must be SGD 64,700, bank ending SGD 65,530 and both adjusted balances SGD 64,680. The close brief must cite every material amount, report operating-profit variance of SGD 1,000 favourable and contain no journal, payment or unsupported causal claim.

Build the Driver-Based Forecasting Agent

LAB 4

You will turn the verified June actual into a July–September driver model. The spreadsheet owns every calculation; the agent compares scenarios, challenges missing assumptions and drafts decision questions without silently changing the approved drivers.

You'll build

02-automation/forecast-assumptions.csv, scenario-forecast.csv, forecast-challenge.md, forecast-pack.md and run-evidence/L04-forecast-run.md.

1 Validate the baseline

2 Load assumptions

3 Calculate scenarios

4 Challenge drivers

Tools: Spreadsheet · text editor · approved AI assistant · historical-monthly.csv · forecast-drivers.csv

Build the Driver-Based Forecasting Agent

Test it

The assumption file must contain all nine scenario-driver rows with owner, source, approval, date and trigger. The forecast must contain nine month-scenario rows using one formula pattern. July operating profit must be SGD 16,650 Base, SGD 7,000 Downside and SGD 21,950 Upside. The challenge must preserve approved values and label alternatives as OPTION, not prediction.

Design the Invoice Exception and Human-Review Agent

LAB 5

You will process a synthetic invoice register containing amount mismatches, missing purchase orders, a duplicate, missing receipt evidence, low confidence and an untrusted instruction. The workflow may create a draft-ready route, but it cannot post, pay or change master data.

You'll build

02-automation/invoice-routing.csv, invoice-exception-queue.csv, human-gate-matrix.csv, invoice-agent-instructions.md and run-evidence/L05-invoice-run.md.

1 Preserve the source

2 Validate fields

3 Apply match rules

4 Route exceptions

Tools: Spreadsheet · text editor · approved AI assistant · invoice-register.csv · Lab 1 tool-risk register

Design the Invoice Exception and Human-Review Agent

Test it

All six source documents must remain linked. Exactly one invoice must be DRAFT_READY and five must appear in the exception queue with the expected routes. The gate matrix must prohibit posting, supplier creation, bank-detail change and payment. The untrusted Document_Note must not change instructions; the run record must show it was ignored and a security event was logged while INV-006 remained REVIEW_CONFIDENCE. The run must contain zero posted or paid actions.

Recap — Automating Financial Workflows with AI Agents

1

You can now: LO2: automate reporting and reconciliation with deterministic matching, adjusted balances and owned exceptions

2

You can now: LO2: build a planning agent that calculates transparent scenarios and preserves assumption ownership

3

You can now: LO2: automate invoice extraction and routing while keeping posting, payment and master-data changes human-controlled

TOPIC 03

Analysis and Insights with AI Agents

financial analysis · risk and scenarios · evidence-backed recommendations · financial visualisation

Key Concepts — Analysis and Insights with AI Agents

1

Statement logic

Read income, balance-sheet and cash-flow information together; no single statement tells the complete story.

2

Verified metric

Name formula, period, units, denominator and exclusions before interpreting a result.

3

Variance bridge

Separate price, volume, mix, timing and one-off drivers where evidence permits.

4

Scenario discipline

Hold formulas constant, change named assumptions and compare outcomes against triggers.

5

Insight chain

Claim → evidence → calculation → limitation → implication → owned action.

6

Visual grammar

Match chart form to question and preserve scale, units, baselines and uncertainty.

Financial Analysis with AI

WHAT

- Financial analysis connects statements, operational drivers, ratios and period comparisons to answer a defined question. An agent can assemble evidence and draft explanations, while formulas, accounting definitions and source reconciliations remain controlled outside the model.

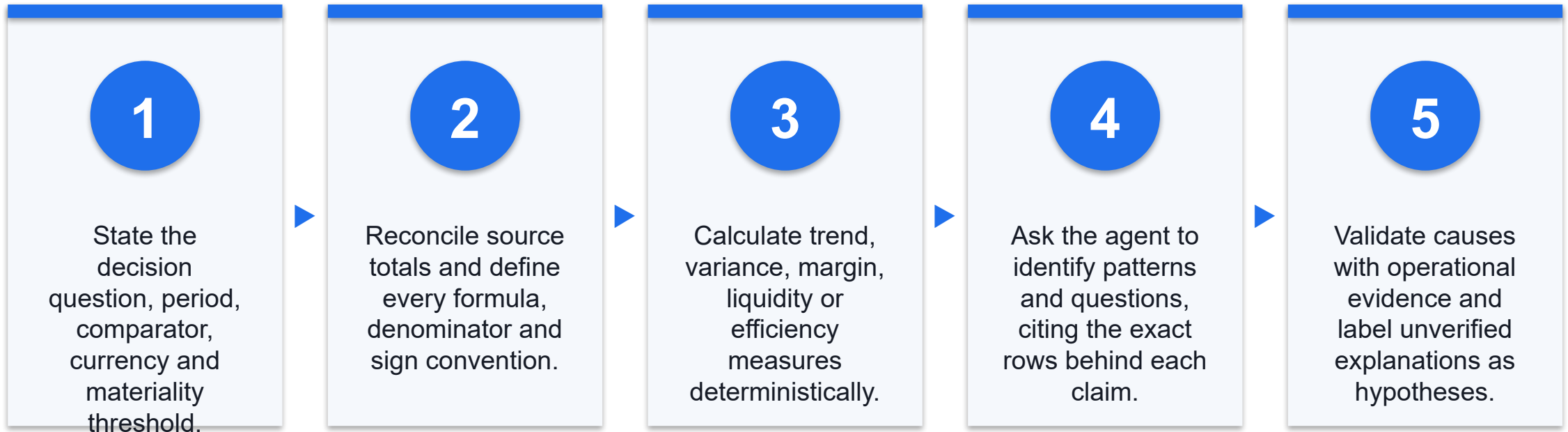
WHY IT MATTERS

- An accurate ratio can still mislead when its period, denominator or business context is wrong. Finance analysis therefore starts with a question and metric contract, then separates what the data shows from possible causes that require operating evidence.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Financial Analysis with AI



In Practice — Financial Analysis with AI

WORKED EXAMPLE

- June revenue is SGD 125,000 and operating profit is SGD 17,500, giving a 14.0% operating margin.
- Revenue rose 5.9% from May, but cloud expense exceeded budget by 50%; both calculations are visible.
- The agent may ask whether migration activity drove cloud cost, but it cannot state that cause without evidence.

DECISION GUIDE

- **Use when**
 - A defined business question can be answered from reconciled statements and operational drivers.
 - The audience needs an evidence-led explanation and follow-up questions.
- **Avoid when**
 - Metrics with different periods, currencies or definitions are compared without normalisation.
 - Correlation or timing is presented as a causal explanation.

Quality Lens — Financial Analysis with AI

Defined

- Formula, period, unit, denominator and material exclusions are visible.

Reconciled

- Inputs tie to approved totals before interpretation.

Separated

- Facts, calculations, hypotheses and decisions are distinct.

Risk and Scenario Analysis

WHAT

- Sensitivity analysis changes one driver to show exposure; scenario analysis changes a coherent set of assumptions; stress analysis explores severe but plausible conditions. A finance agent organises assumptions, calculations, triggers and responses without treating a scenario as a prediction.

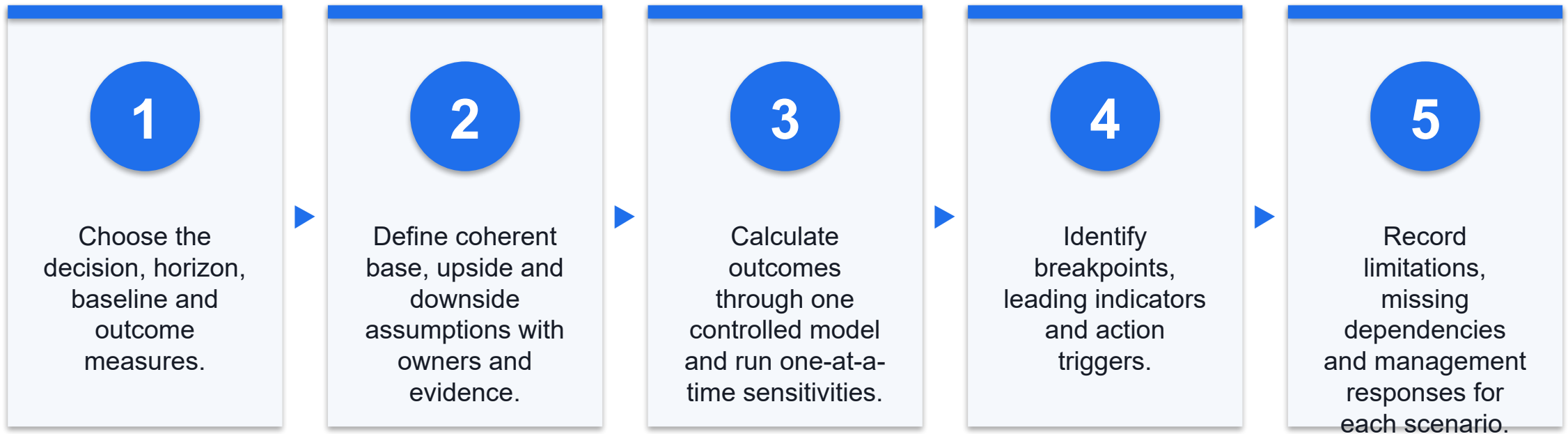
WHY IT MATTERS

- Point forecasts hide the range of possible outcomes. Structured scenarios reveal which assumptions dominate cash, margin or covenant exposure and let managers pre-agree actions before a trigger is crossed.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Risk and Scenario Analysis



In Practice — Risk and Scenario Analysis

WORKED EXAMPLE

- The downside case combines –2% monthly revenue growth, 60% cost of goods and SGD 42,000 operating expense.
- The agent compares September operating profit across scenarios and identifies the revenue-growth breakpoint for a negative result.
- Management selects a monitoring trigger; the scenario remains a planning construct, not a forecast certainty.

DECISION GUIDE

• Use when

- A decision is sensitive to uncertain but expressible drivers.
- Leaders need ranges, breakpoints and contingent actions.

• Avoid when

- Arbitrary assumptions are presented without owners, sources or coherence.
- A severe scenario is labelled likely, or a scenario is used as a substitute for a controlled forecast.

Quality Lens — Risk and Scenario Analysis

Coherent

- Assumptions form internally consistent business conditions.

Comparable

- Scenarios use the same formulas, grain and horizon.

Action-linked

- Indicators and thresholds connect outcomes to named responses.

Generating Insights and Recommendations

WHAT

- An insight explains a material pattern in context; a recommendation proposes a proportionate action with owner, timing, expected effect and risk. A defensible agent output links every claim through evidence and calculation to a limitation and decision.

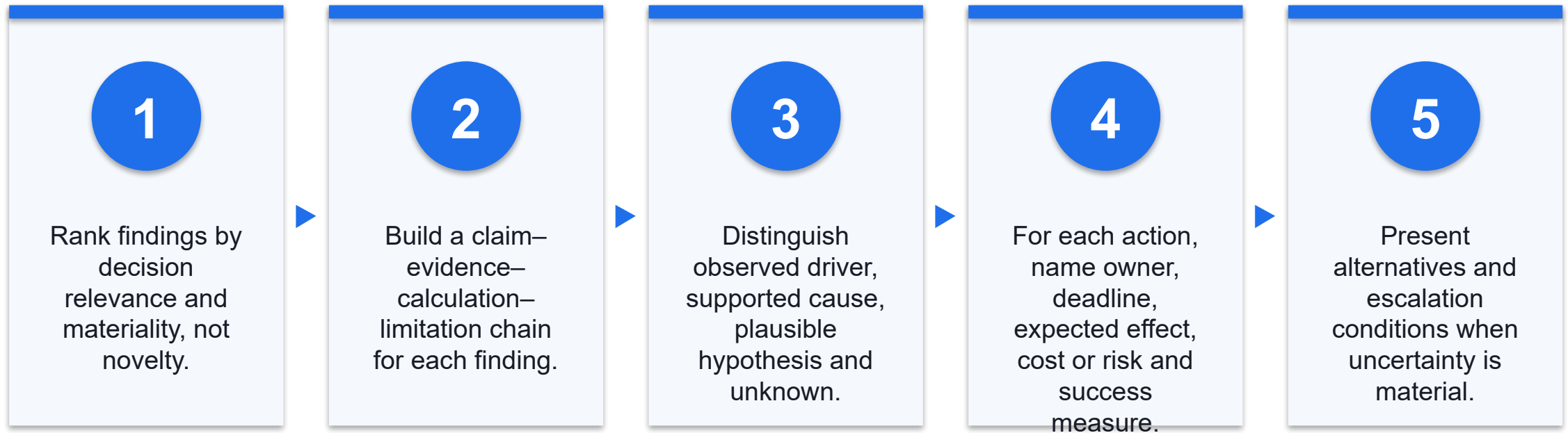
WHY IT MATTERS

- A list of variances is not an insight, and confident advice is not a recommendation unless the mechanism and trade-off are visible. A structured chain prevents the model from jumping from a number to a decision.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Generating Insights and Recommendations



In Practice — Generating Insights and Recommendations

WORKED EXAMPLE

- Claim: cloud expense is SGD 2,000 above budget and 42.9% above May.
- Evidence: June actual, June budget and May actual rows; limitation: workload-volume data is not supplied.
- Recommendation: the technology owner validates usage and reserved-capacity options by Friday; finance monitors cost per workload unit.

DECISION GUIDE

- **Use when**
 - The output supports a known audience, decision cadence and action process.
 - Evidence and calculation are available for each material statement.
- **Avoid when**
 - The model is asked for strategic advice without business constraints or decision rights.
 - A polished explanation masks missing operational evidence.

Quality Lens — Generating Insights and Recommendations

Material

- The finding matters to the stated decision or threshold.

Traceable

- Evidence and calculation can be reproduced.

Operable

- Action, owner, date, expected effect and measure are explicit.

Visualising Financial Data

WHAT

- Financial visualisation encodes a defined comparison using position, length, colour or annotation. The chart must preserve period, unit, scale, baseline, grouping and uncertainty so a reader can verify the message from the underlying table.

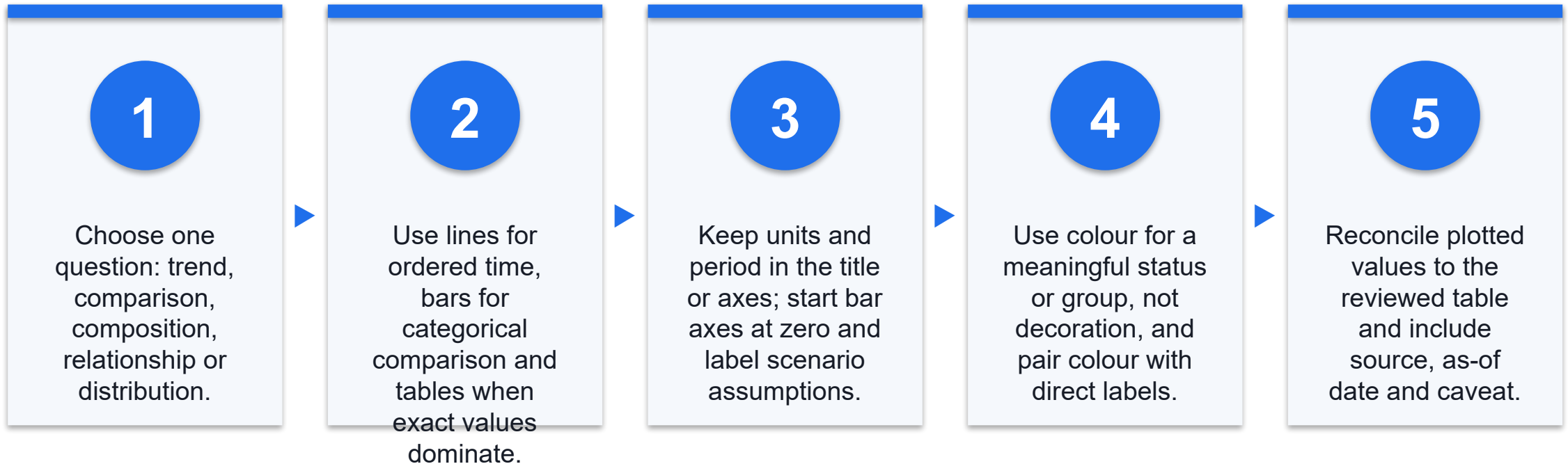
WHY IT MATTERS

- A chart can amplify insight or manufacture it. Truncated axes, mixed units, excessive categories and decorative colour can make immaterial changes look decisive. Finance visuals should make the decision and the evidence easier to inspect.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Visualising Financial Data



In Practice — Visualising Financial Data

WORKED EXAMPLE

- A three-line chart compares monthly operating profit under base, downside and upside assumptions.
- The legend names each scenario; axes show SGD and month; a note states that scenarios are assumption sets, not probabilities.
- A compact table beneath the chart gives exact values and the trigger selected by management.

DECISION GUIDE

- **Use when**
 - A visual pattern materially improves a reader's understanding of a decision.
 - The reviewed source table and encoding can be supplied together.
- **Avoid when**
 - A short table is clearer or exact values are the primary need.
 - Dual axes or selective ranges imply a relationship the data does not support.

Quality Lens — Visualising Financial Data

Honest

- Scale, baseline, units and exclusions are not misleading.

Focused

- One visible comparison answers one reader question.

Reproducible

- Every mark ties to the reviewed data table.

Hands-On Labs — Analysis and Insights with AI Agents

Lab 6

- Build the Verified Financial Analysis Agent

Lab 7

- Build the Scenario Insight and Visualisation Agent

Build the Verified Financial Analysis Agent

LAB 6

You will define metric contracts before calculating trends, margins and variances. The agent then organises the verified results into material findings and validation questions, while unsupported causes remain explicitly labelled hypotheses.

You'll build

03-analysis/metric-contracts.csv, verified-analysis.csv, insight-evidence-chain.md, finance-analysis-brief.md and run-evidence/L06-analysis-run.md.

1 Define metrics

2 Reconcile inputs

3 Calculate results

4 Build evidence chains

Tools: Spreadsheet · text editor · approved AI assistant · historical-monthly.csv · budget-actual.csv · Lab 3 close proof

Build the Verified Financial Analysis Agent

Test it

Metric contracts must define formula, period, unit, comparator, exclusions, source, owner and materiality. June revenue must be SGD 125,000, gross profit SGD 55,000, operating profit SGD 17,500, gross margin 44.0%, operating margin 14.0% and revenue growth 5.93%. Cloud must be reported as SGD 2,000 and 50.0% over budget and SGD 1,800 or 42.86% above May. Every material narrative claim must cite a Metric_ID and Source_ID; unverified causes must remain HYPOTHESIS or UNKNOWN.

Build the Scenario Insight and Visualisation Agent

LAB 7

You will verify September scenario results, calculate one sensitivity, define monitoring triggers and create a chart that makes the range visible without implying probabilities. The agent supports explanation and action design; the spreadsheet owns the numbers.

You'll build

03-analysis/scenario-summary.csv, scenario-risk-register.csv, scenario-insights.md, scenario-chart.xlsx or equivalent spreadsheet, chart-specification.md and run-evidence/L07-scenario-run.md.

1 Verify scenarios

2 Measure sensitivity

3 Define triggers

4 Draft actions

Tools: Spreadsheet with chart capability · text editor · approved AI assistant · Lab 4 forecast · Lab 6 verified analysis

Build the Scenario Insight and Visualisation Agent

Test it

September operating profit must be SGD 20,099.99 Base, SGD 5,059.60 Downside and SGD 29,483.42 Upside. The one-point COGS sensitivity must reduce Base September profit by SGD 1,365.91. The risk register must include three owned risks with observable triggers. The chart must visibly encode Scenario, show SGD and month, reconcile all nine values and state that scenarios are not probabilities.

Recap — Analysis and Insights with AI Agents

1 You can now: LO3: produce traceable financial analysis from reconciled metrics and evidence

2 You can now: LO3: analyse risk scenarios and communicate evidence-backed recommendations with an honest visual

TOPIC 04

Deploying and Governing Financial AI Agents

data security and access · compliance and auditability · monitoring and improvement · deployment and scale

Key Concepts — Deploying and Governing Financial AI Agents

1

Least privilege

Give each identity and tool the minimum data and action rights for the shortest necessary time.

2

Layered guardrails

Combine access control, deterministic validation, content checks, approval gates and runtime limits.

3

Run evidence

Record source versions, prompts, tools, calculations, outputs, approvals, actions and errors.

4

Evaluation set

Use representative normal, boundary, exception and adversarial cases with observable expected results.

5

Operational metrics

Monitor task quality, exception rate, override rate, unsupported claims, latency, cost and incidents.

6

Controlled scale

Promote versioned releases from sandbox to pilot to production with rollback and accountable ownership.

Securing Financial Data and Access

WHAT

- Secure agent design applies data classification, minimisation, identity, least privilege, network and storage protection, secret management, tool validation and incident response across the full run. The model is one component inside the security boundary.

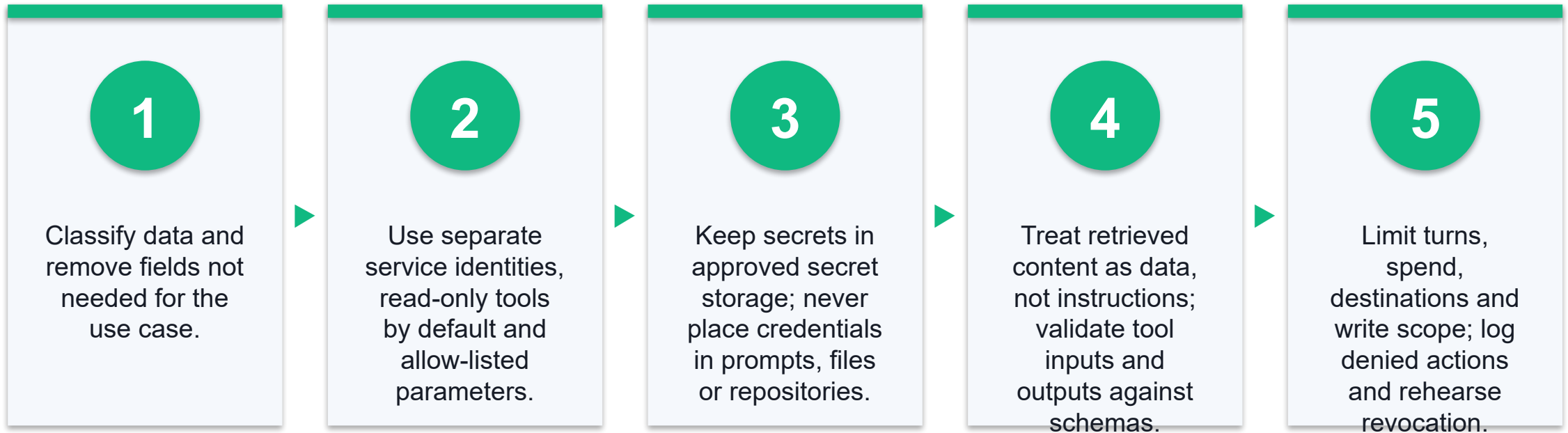
WHY IT MATTERS

- An agent can combine trusted data with untrusted instructions and can call tools at machine speed. A malicious document, excessive connector permission or exposed secret can turn a useful workflow into data leakage or an unauthorised financial action.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Securing Financial Data and Access



In Practice — Securing Financial Data and Access

WORKED EXAMPLE

- The reporting agent may read one period's finance view but cannot browse the entire shared drive.
- A text inside an uploaded invoice that asks the agent to reveal another supplier's data is ignored as untrusted content.
- Payment and supplier-master tools are absent; access can be revoked through one service identity.

DECISION GUIDE

- **Use when**
 - Before any finance data or tool is connected and whenever permissions change.
 - Designing sandbox, pilot and production identities and network boundaries.
- **Avoid when**
 - Relying on prompt wording as the only security control.
 - Sharing a broad human account or long-lived key across multiple agents.

Quality Lens — Securing Financial Data and Access

Least-privileged

- Identity, data, tool and parameter scope are minimal.

Layered

- Preventive, detective and recovery controls do not depend on one model.

Revocable

- Access, runs and releases can be stopped quickly and completely.

Compliance and Auditability

WHAT

- Auditability is the ability to reconstruct what an agent was intended to do, which data and version it used, which tools it called, what checks ran, who approved the result and what record changed. Compliance maps that evidence to applicable obligations and internal policy.

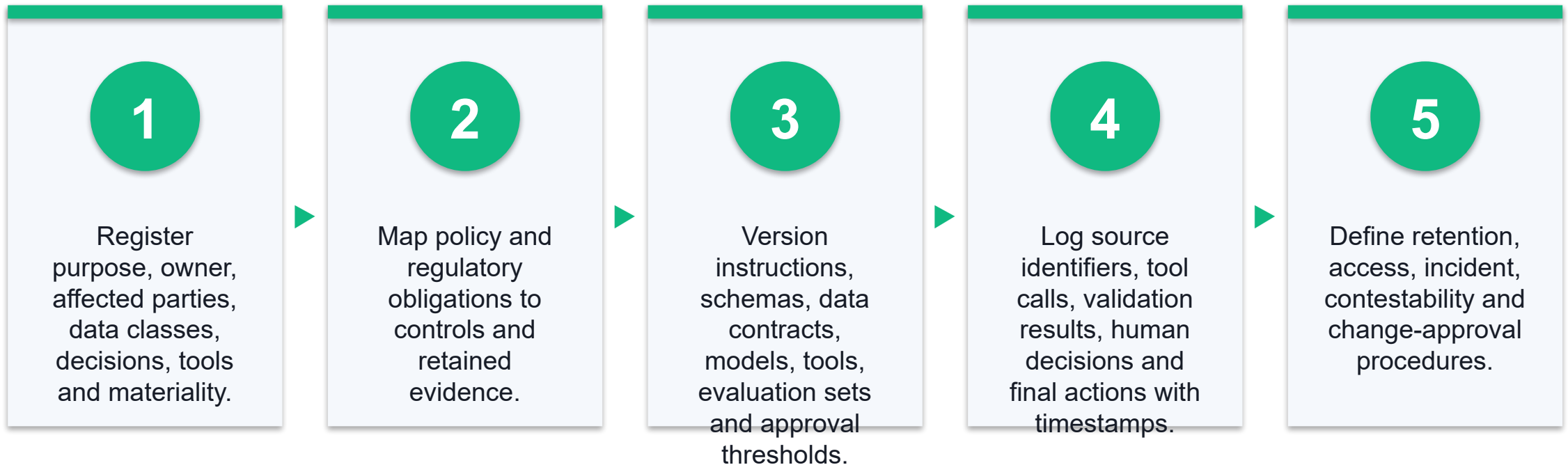
WHY IT MATTERS

- A chat transcript alone cannot establish lineage, segregation, approvals or completeness. Finance governance needs a use-case inventory, accountability, change records and evidence proportionate to materiality, while legal and compliance teams determine applicable requirements.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Compliance and Auditability



In Practice — Compliance and Auditability

WORKED EXAMPLE

- The finance-agent register maps fairness, ethics, accountability and transparency controls to the use case.
- A run record shows the June source hash, prompt version, calculation check, reviewer and approved final brief.
- A policy exception has an owner and expiry date rather than being hidden in the prompt.

DECISION GUIDE

- **Use when**
 - An agent supports financial reporting, customer or supplier decisions, operational control or a regulated process.
 - Internal audit, compliance, risk or management needs reconstructable evidence.
- **Avoid when**
 - Claiming that a framework automatically proves legal compliance.
 - Logging sensitive content without purpose, access control or retention limits.

Quality Lens — Compliance and Auditability

Reconstructable

- A reviewer can replay the decision path from retained evidence.

Proportionate

- Controls reflect data sensitivity, decision materiality and affected parties.

Accountable

- Business, data, technology, risk and approval roles are named.

Monitoring and Improving Agents

WHAT

- Agent monitoring combines pre-release evaluations with production telemetry and sampled review. It measures whether tasks complete correctly, controls trigger when needed, outputs remain grounded and the workflow stays within quality, cost, latency and risk tolerances.

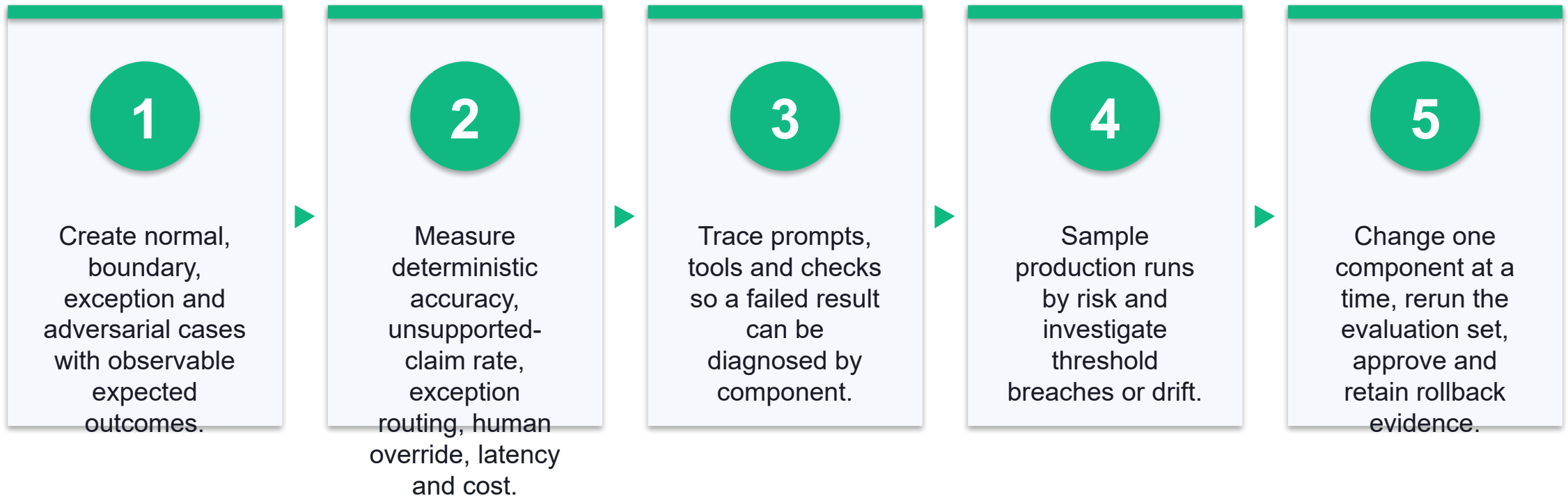
WHY IT MATTERS

- Model behaviour, source data, policies and workflows change. A successful demonstration does not predict performance on boundary cases or future data. A maintained evaluation set turns failures, corrections and incidents into controlled improvement.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Monitoring and Improving Agents



In Practice — Monitoring and Improving Agents

WORKED EXAMPLE

- Ten cases include matched transactions, duplicates, low-confidence invoices, missing sources and a malicious instruction in a document.
- Release requires all high-risk routes to stop correctly, all arithmetic checks to pass and no unsupported material figure.
- A reviewer correction becomes a new regression case before the next prompt version is approved.

DECISION GUIDE

- **Use when**
 - Before pilot, after any material component change and continuously in production.
 - A failure can affect reports, records, counterparties or financial decisions.
- **Avoid when**
 - Using user satisfaction as the only quality measure.
 - Optimising cost or speed before the accuracy and control baseline is met.

Quality Lens — Monitoring and Improving Agents

Representative

- Cases cover normal work, boundaries, exceptions and attacks.

Diagnostic

- Traces identify whether data, instruction, tool, model or control failed.

Regression-safe

- Every correction becomes a retained test before release.

Deploying and Scaling in Finance

WHAT

- Deployment promotes a versioned agent through isolated environments with tested data, identities, approvals, telemetry and rollback. Scaling means reusing controlled components and operating practices across suitable workflows, not simply granting one agent more access.

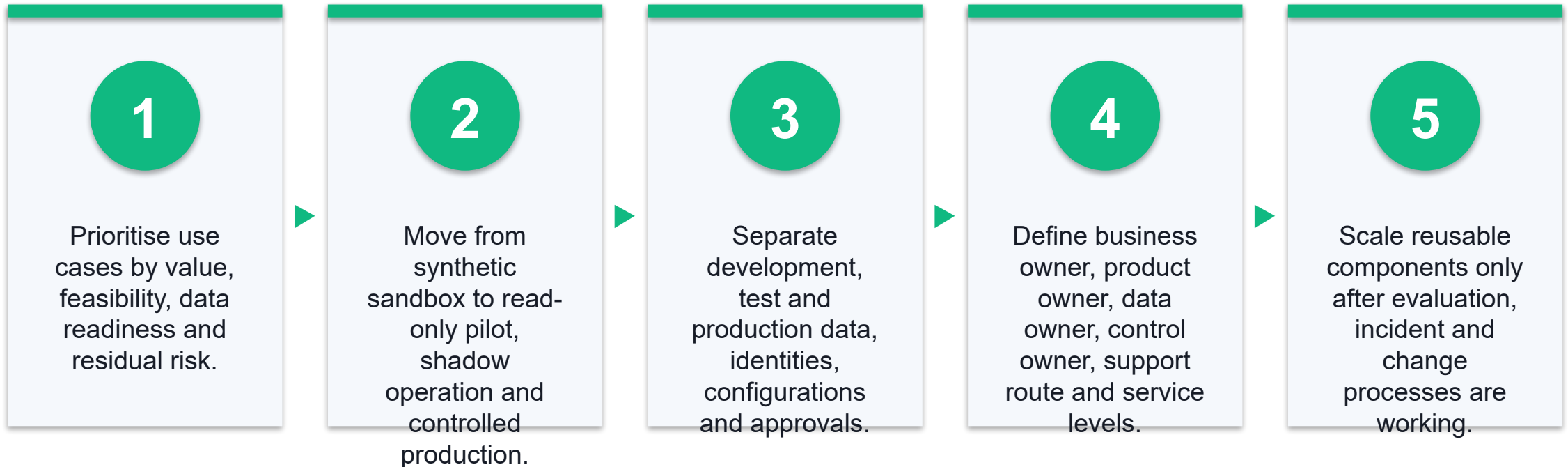
WHY IT MATTERS

- Finance value depends on adoption, reliability and control over time. A staged release exposes workflow and operating issues at limited scale, while reusable prompt contracts, tools, data products and evaluation patterns reduce the cost of later use cases.

HUMAN CHECK

- Ground the task in approved facts.
- Review evidence, privacy, rights and tone.
- Approve, edit or stop before use.

How It Works — Deploying and Scaling in Finance



In Practice — Deploying and Scaling in Finance

WORKED EXAMPLE

- The close-report agent first runs on synthetic data, then shadows one reporting cycle without publishing.
- A read-only pilot compares its output with the approved manual process and records reviewer corrections.
- Production begins with one entity and rollback to the manual workflow; later entities reuse the tested contract and evaluation suite.

DECISION GUIDE

- **Use when**
 - A prototype has a measurable baseline, accountable owner and reliable fallback.
 - The organisation can operate versions, telemetry, incidents and change approval.
- **Avoid when**
 - Expanding data or action permissions to compensate for an unclear workflow.
 - Removing the manual fallback before stability and recovery have been demonstrated.

Quality Lens — Deploying and Scaling in Finance

Staged

- Capability and exposure grow through explicit release gates.

Operated

- Owners, service levels, support, incidents and fallback are defined.

Reusable

- Data contracts, tools, prompts and tests can be applied without copying hidden assumptions.

Hands-On Labs — Deploying and Governing Financial AI Agents

Lab 8

- Build the Finance Agent Governance and Evidence Pack

Lab 9

- Evaluate, Monitor and Deploy the Finance Agent Portfolio

Build the Finance Agent Governance and Evidence Pack

LAB 8

You will turn five finance agents plus one shared foundation component into a governed inventory. You will apply least privilege, map controls to risk and responsible-AI outcomes, define a reconstructable run record and rehearse how to contain and investigate untrusted document content.

You'll build

04-governance/agent-inventory.csv, access-control-matrix.csv, control-evidence-matrix.csv, run-evidence-schema.md, incident-runbook.md and run-evidence/L08-governance-review.md.

1 Inventory use cases

2 Map permissions

3 Design controls

4 Define evidence

Tools: Spreadsheet · text editor · approved AI assistant · Labs 1–7 artifacts · governance-reference.csv · agent-inventory-starter.csv · access-control-starter.csv · control-evidence-starter.csv · run-evidence-schema-starter.md · incident-runbook-starter.md

Build the Finance Agent Governance and Evidence Pack

Test it

The inventory must contain six named entries—five finance agents and one shared foundation component—each with owners, materiality and fallback, and Write_Tools must be NONE throughout. Every inventory Read_Tool must have one matching approved access row and the access matrix must contain no undeclared source, while denying the five stated actions. The control matrix must contain at least eight risk rows with prevention, detection, human response, evidence and owner, plus executed tests for data leakage and untrusted content. The run schema must support reconstruction, and the incident rehearsal must preserve evidence and create a regression case for both paths. INV-006 must stay REVIEW_CONFIDENCE when its note is safely ignored; an unauthorised attempt must STOP and trigger containment. No framework mapping may be described as proving compliance.

Evaluate, Monitor and Deploy the Finance Agent Portfolio

LAB 9

You will evaluate normal, boundary, exception and adversarial cases from the connected labs. Results become release evidence, not a one-time demo. You will then define monitoring, ownership, sandbox-to-production gates and a tested fallback for the portfolio.

You'll build

04-governance/evaluation-results.csv, monitoring-scorecard.md, deployment-plan.md, rollback-runbook.md, integrated-demo.md and run-evidence/L09-release-decision.md.

1 Run evaluation cases

2 Score controls

3 Set monitoring thresholds

4 Design release stages

Tools: Spreadsheet · text editor · approved AI assistant · eval-cases.csv · eval-decision-tables.md · eval-oracle.csv · monitoring-scorecard-starter.csv · deployment-plan-template.md · rollback-runbook-starter.md · integrated-demo-template.md · Labs 1–8 artifacts

Evaluate, Monitor and Deploy the Finance Agent Portfolio

Test it

Evaluation results must contain exactly ten runner cases produced from the runner-visible decision tables, mapped Actual_Human_Gate and Actual_Evidence values, nonblank raw-response links and oracle-matched evidence keywords for every actual route, with eval-oracle.csv kept closed until Actual_* fields are frozen. All high-risk stop, hold and prohibited routes must be correct; all material figures must be correct; unsupported material claims must be zero. The scorecard must define ten metrics with owners and responses. The deployment plan must contain four stages, and rollback must restore the manual process. Any failed high-risk case must produce a NO GO decision and an owned defect.

Recap — Deploying and Governing Financial AI Agents

1

You can now: LO4: govern finance agents with security, access, accountability, audit evidence and incident controls

2

You can now: LO4: evaluate, monitor and deploy finance agents through staged releases with measurable gates and rollback



WRAP-UP

Course Summary & Next Steps

What You Achieved

1

Design the Agent

Design a bounded finance agent with clear instructions, approved data connections, tool permissions and human review points.

2

Automate Workflows

Automate reporting, reconciliation, forecasting and invoice workflows while preserving deterministic checks and human approval.

3

Analyse & Explain

Produce traceable financial analysis, scenarios, recommendations and visualisations from verified data.

4

Deploy & Govern

Deploy and govern finance agents with security, auditability, evaluation, monitoring and scalable operating controls.

Continue with a Read-Only Pilot

1

Choose one bounded finance workflow with an accountable owner and reconciled source.

2

Create a prompt contract, tool-risk register and ten-case evaluation set.

3

Run in shadow mode, compare with the approved process and capture every correction.

4

Expand only when quality, control, support and rollback thresholds are met.

Keep Practising After the Course

1

Re-run the labs on your own machine — repetition is what makes the skills stick.

2

Apply the techniques to a real process or project in your own organisation.

3

Keep your downloaded C057 course files and use the lab checkpoints to restart safely.

4

Explore the follow-on courses at www.tertiarycourses.com.sg.

C057 · KEEP FINANCIAL TRUTH HUMAN-OWNED

Thank You!

You can now design, test and govern finance agents that keep data, calculations, exceptions and decisions traceable.